

Airbnb Price Prediction

Final Machine Learning Project | ITAI 1371 | Mary Balemba

Goal: Predict Airbnb listing price using regression models.

Target: price (numeric) → Regression problem.

Dataset: Airbnb Open Data; cleaned dataset used for final modeling.

Required split: 70% Train / 15% Validation / 15% Test.

Clean Data

Remove Leakage

Train 5 Models

Ensemble + Bayesian

Test & Select

Methodology and Data Correction

Models: Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and KNN.

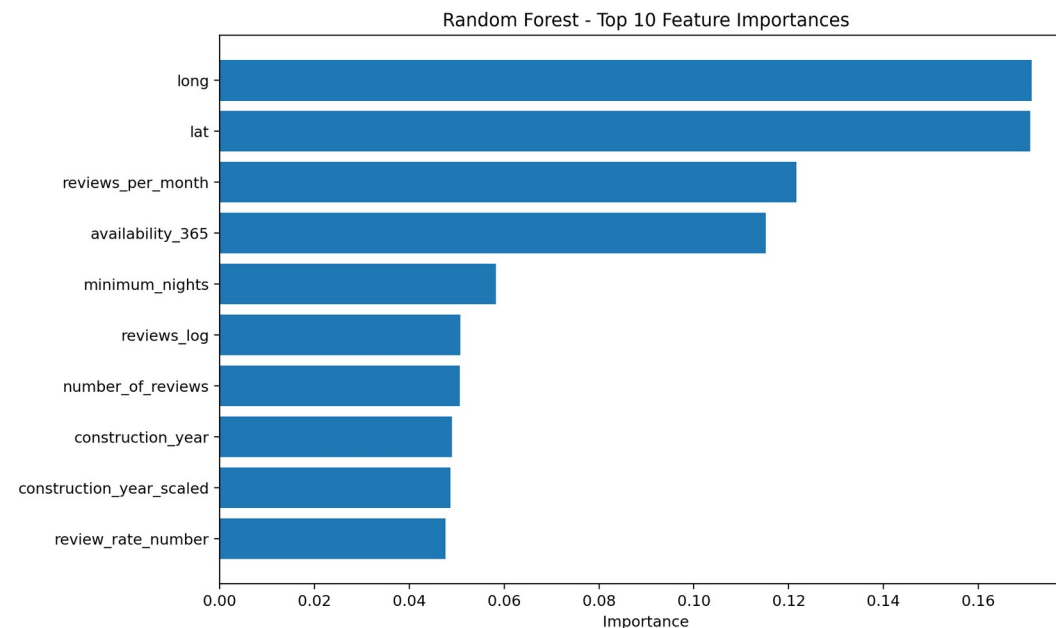
Metrics: MAE, MSE, and R^2 on Validation and Test sets.

Average ensemble used predictions from the top 3 validation models.

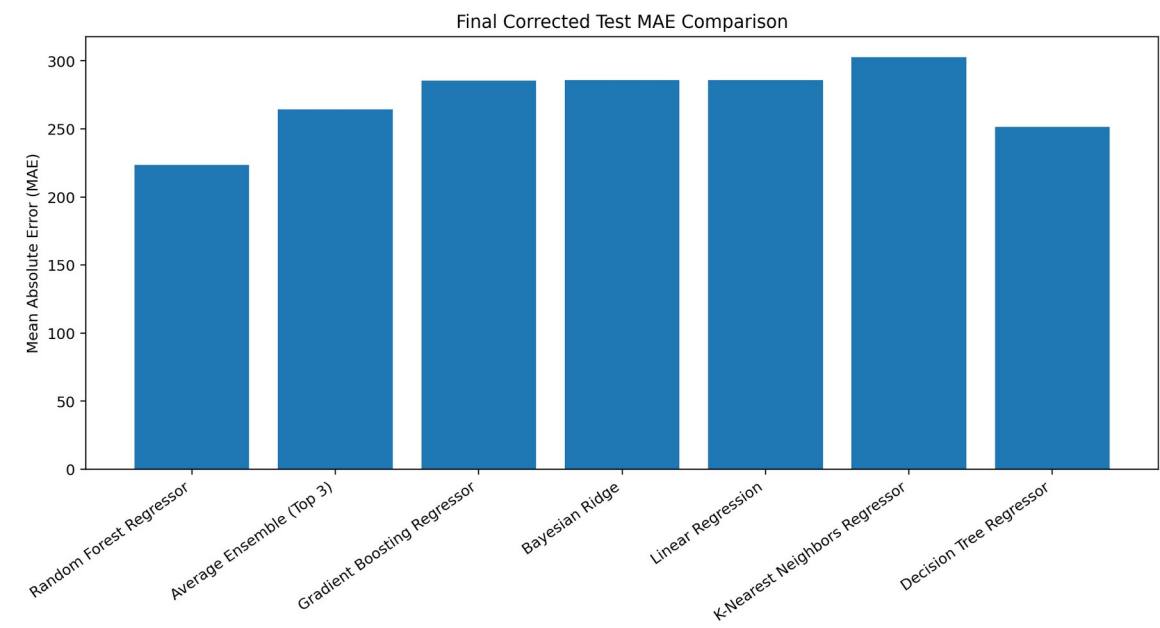
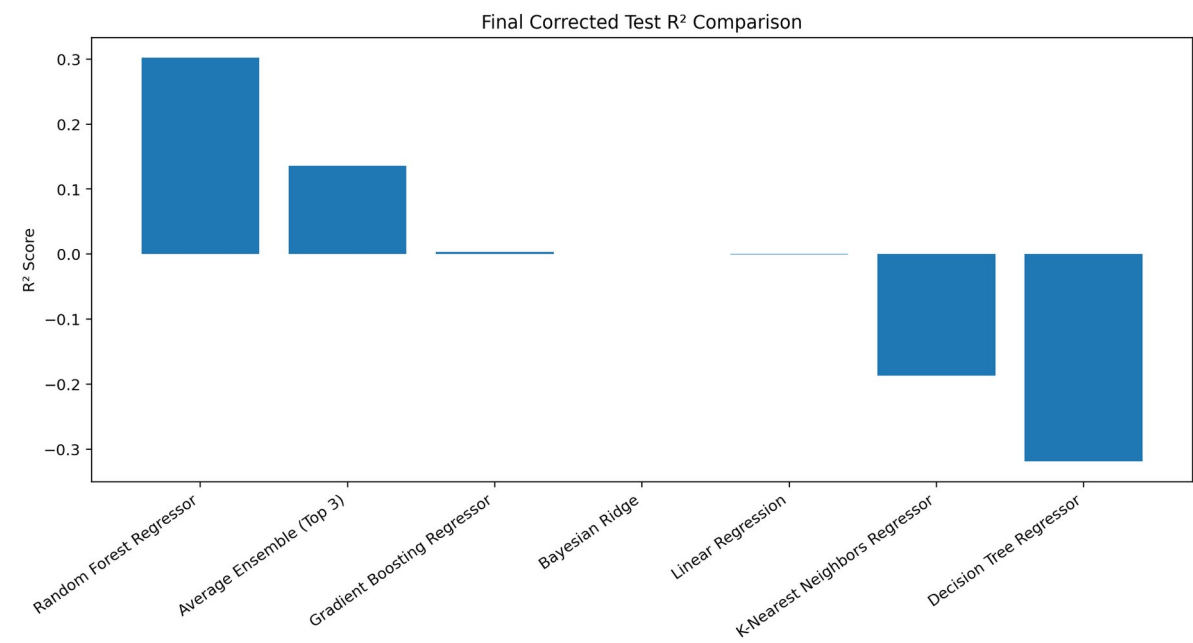
Bayesian Ridge used for the Bayesian comparison.

Sanity check found target leakage: service_fee correlated ≈ 0.998 with price; fee_to_price_ratio was price-derived.

Both leakage variables were removed and every model was retrained.



Final Model Results



Winner: Random Forest — Test MAE 223.6204 | Test MSE 76,230.0658 | Test R² 0.3022

Conclusion and Lessons Learned

Random Forest was the best model: highest Test R^2 and lowest Test MAE/MSE.

Test $R^2 = 0.3022 \rightarrow$ the model explained about 30.22% of price variance.

Validation R^2 (0.3031) and Test R^2 (0.3022) were very close, showing stable generalization.

The Top-3 Average Ensemble ranked second but did not outperform Random Forest.

Main lesson: unusually high scores must be checked for target leakage before accepting the result.

Future work: richer location/amenity features, cross-validation, tuning, XGBoost/LightGBM.

Final selected model: Random Forest Regressor